

PET and Gestational Hypertension 10: Integrated Cases and Final Synthesis

Companion reading handout for simultaneous listening.

Generated from the final TTS transcript for episode 10. Voice: Kore.

This final episode is about using the language under pressure.

The goal is not to recite definitions. The goal is to hear a case and immediately ask the right next question.

Case One

Case one.

A thirty eight week patient has new blood pressure readings of one hundred forty eight over ninety four and one hundred fifty over ninety two. She has no headache, no visual symptoms, normal platelets, normal creatinine, normal liver enzymes, and no proteinuria. The fetus is reassuring.

The diagnosis is gestational hypertension without severe features at term.

The management is delivery, usually induction unless there is another obstetric reason. The reason is HYPITAT logic. At thirty seven weeks or beyond, continuing the pregnancy keeps accumulating maternal risk while fetal maturity benefit is small.

The trap is to say, no protein, so she can wait. That is old thinking.

Case Two

Case two.

A thirty one week patient has new gestational hypertension. Protein is negative. Labs are normal. But the sFlt-1 to PlGF ratio is high, and the fetus is beginning to fall across growth percentiles.

This is not a patient for casual reassurance. Depending on the guideline, she may still be labeled gestational hypertension or may meet a broader uteroplacental dysfunction definition of pre-eclampsia. Either way, the biology is concerning. A high ratio is not diagnostic by itself, but it suggests placental antiangiogenic disease. Growth deceleration suggests placental dysfunction.

The next move is close maternal and fetal surveillance, not delivery just because the biomarker is high, and not discharge as if the case is simple.

The trap is to treat biomarkers as either magic diagnosis or meaningless noise. They are neither.

Case Three

Case three.

A patient with chronic hypertension and baseline proteinuria presents at twenty nine weeks. Her BP is worsening. Proteinuria has increased. Creatinine is rising from baseline. Platelets are falling but still above one hundred thousand.

This is possible superimposed pre-eclampsia. The diagnosis is difficult because chronic kidney disease and chronic hypertension create baseline abnormalities. You need trend, symptoms, labs, fetal growth, Doppler, and sometimes angiogenic testing.

The trap is to use proteinuria alone. In chronic renal disease, proteinuria can rise during pregnancy. The more important question is whether new organ dysfunction or uteroplacental dysfunction has appeared.

Case Four

Case four.

A thirty five week patient has pre-eclampsia, BP one hundred fifty over ninety six, normal platelets, normal creatinine, mildly elevated proteinuria, no symptoms, and reassuring fetal status.

This is late-preterm pre-eclampsia without severe features. The decision is nuanced. ACOG severe-feature rules do not mandate delivery because severe features are absent. NICE and ISSHP discuss planned birth in this window based on risk, resources, steroids, and patient values. PHOENIX teaches that planned delivery reduces maternal morbidity and severe hypertension but increases neonatal unit admission.

The teaching phrase is: delivery buys maternal safety and spends neonatal maturity.

The trap is to pretend thirty five weeks is the same as thirty seven, or the same as thirty one. It is neither.

Case Five

Case five.

A patient has BP one hundred forty six over ninety two, severe right upper quadrant pain, nausea, platelets eighty five thousand, AST and ALT more than twice normal, and LDH elevated.

This is HELLP syndrome, even if the blood pressure is not severe. She has a microangiopathic liver and platelet disease. The correct frame is maternal stabilization and delivery planning, not outpatient observation because the systolic BP is below one hundred sixty.

The trap is to let blood pressure dominate the whole diagnosis. Pre-eclampsia is endothelial disease. Severe organ involvement can be present without severe-range BP.

Case Six

Case six.

A postpartum patient, day five, returns with severe headache and BP one hundred seventy over one hundred ten.

This is a postpartum hypertensive emergency until proven otherwise. Treat severe BP. Evaluate neurologic symptoms. Consider magnesium if pre-eclampsia with severe features or eclampsia risk is present. Check platelets, creatinine, liver enzymes, urine, and symptoms. Do not reassure her because the placenta is out.

The trap is to think delivery ended the risk. Postpartum days three to seven are a BP peak window.

Case Seven

Case seven.

A normotensive thirty two week patient has severe fetal growth restriction, oligohydramnios, and abnormal umbilical artery Doppler.

This may not be pre-eclampsia under ACOG if maternal hypertension is absent. But it is placental disease and belongs in the same biologic neighborhood. Under ISSHP logic, uteroplacental dysfunction is central to the hypertensive disorder framework, but hypertension remains important for the PET label.

The trap is to ignore the placenta because maternal BP is normal. The fetus may be declaring placental insufficiency before the mother declares hypertension.

Final Course Synthesis

Now let us synthesize the course.

The first anchor is the layered diagnosis.

Hypertension is the entry point. Organ involvement separates isolated gestational hypertension from pre-eclampsia. Severe features identify maternal danger. Gestational age determines whether time is useful or harmful.

The second anchor is the endothelial syndrome.

Pre-eclampsia begins in many cases with abnormal placentation and angiogenic imbalance. sFlt-1 rises, PlGF falls, endothelial health deteriorates. The endothelium explains the kidney, the liver, the platelets, the lungs, the brain, and the placenta. If you understand that, the disease stops being a memorized list.

The third anchor is that proteinuria is not the master variable.

Proteinuria helps diagnose pre-eclampsia. It does not stage severity once the diagnosis is made. Severe features, maternal trajectory, fetal status, and gestational age drive management.

The fourth anchor is maternal safety.

Severe BP is a stroke problem. Magnesium is a seizure prevention drug. Fluids can cause pulmonary edema. HELLP can present without dramatic BP. Eclampsia requires maternal stabilization before fetal panic.

The fifth anchor is delivery timing.

At thirty seven weeks without severe features, delivery is supported by HYPITAT. At thirty four weeks or beyond with severe features, stabilize and deliver. Before thirty four weeks, expectant management is only for carefully selected stable patients in the right center. From thirty four to thirty six plus six, PHOENIX teaches the tradeoff: maternal benefit, more neonatal admission, shared decision-making when stable.

The sixth anchor is postpartum risk.

The placenta is delivered, but the vascular disease can persist. BP often peaks days three to seven. Stroke and eclampsia can occur postpartum. Renal and proteinuria follow-up matter. Long-term cardiovascular prevention is part of the diagnosis.

Closing Synthesis

Let us end with the full resident reflex.

When a pregnant or postpartum patient has hypertension, do not ask only, is there proteinuria?

Ask: is this isolated hypertension or maternal-placental endothelial disease? Are there severe features? Is the fetus showing placental disease? Is gestational age making time useful or dangerous? Which catastrophe am I preventing right now: stroke,

seizure, pulmonary edema, renal failure, liver rupture, abruption, fetal compromise, or postpartum readmission?

That is the point of the course.

GHTN is hypertension before organ disease is documented. PET is hypertension with maternal, placental, or fetal end-organ disease. Severe features and gestational age drive urgency. And every number, every medication, every delivery threshold, and every follow-up visit exists to prevent a specific clinical failure.